

AUTONOMOUS RC CAR WEEK 2 PRESENTATION

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OUR GOAL AS A TEAM

Create an autonomous vehicle that can navigate through an unstructured environment.

Overall Steps

Step 01

Can it run autonomously without using the remote, with sensors and a microcontroller board to which we can upload code?

Step 02

Can it detect and identify obstacles correctly?

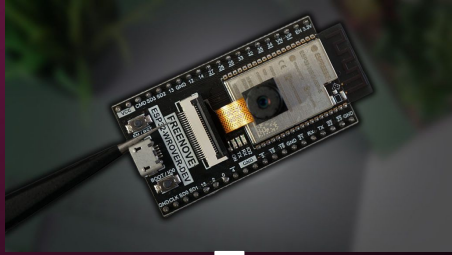
Step 03

Can it apply that obstacle detection to develop a software that can then avoid the obstacles?

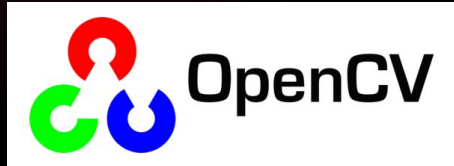
Step 04 (Bonus)

Can it create paths based on given targets?

Assemble Electronics

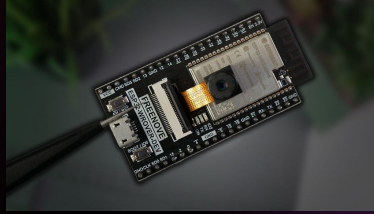


WiFi Video Stream to Laptop

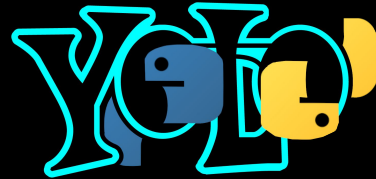


- Receives live camera stream
- Converts video into individual frames

Current Plan



Obstacle Avoidance
Algorithm- Integrate with code

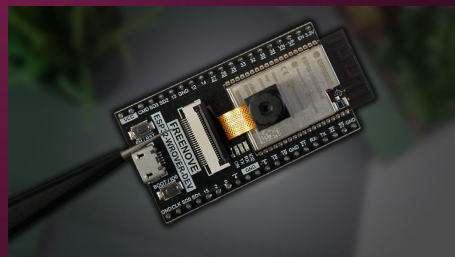


- Analyzes each frame
- Identifies obstacles and determines their location

Basic Steering and Throttle Commands



Possible Future Plan



NVIDIA Jetson Orin (\$250)



Navigation Logic



Distance sensing

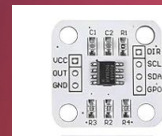
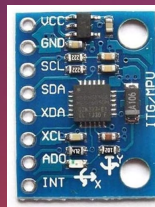


LiDAR Sensor



Relative Location and Movement Sensing

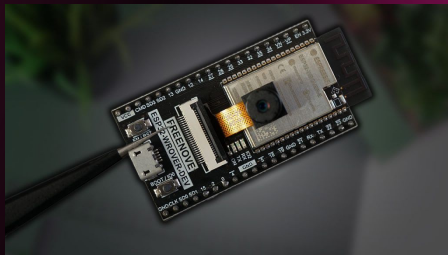
IMU



Wheel encoders

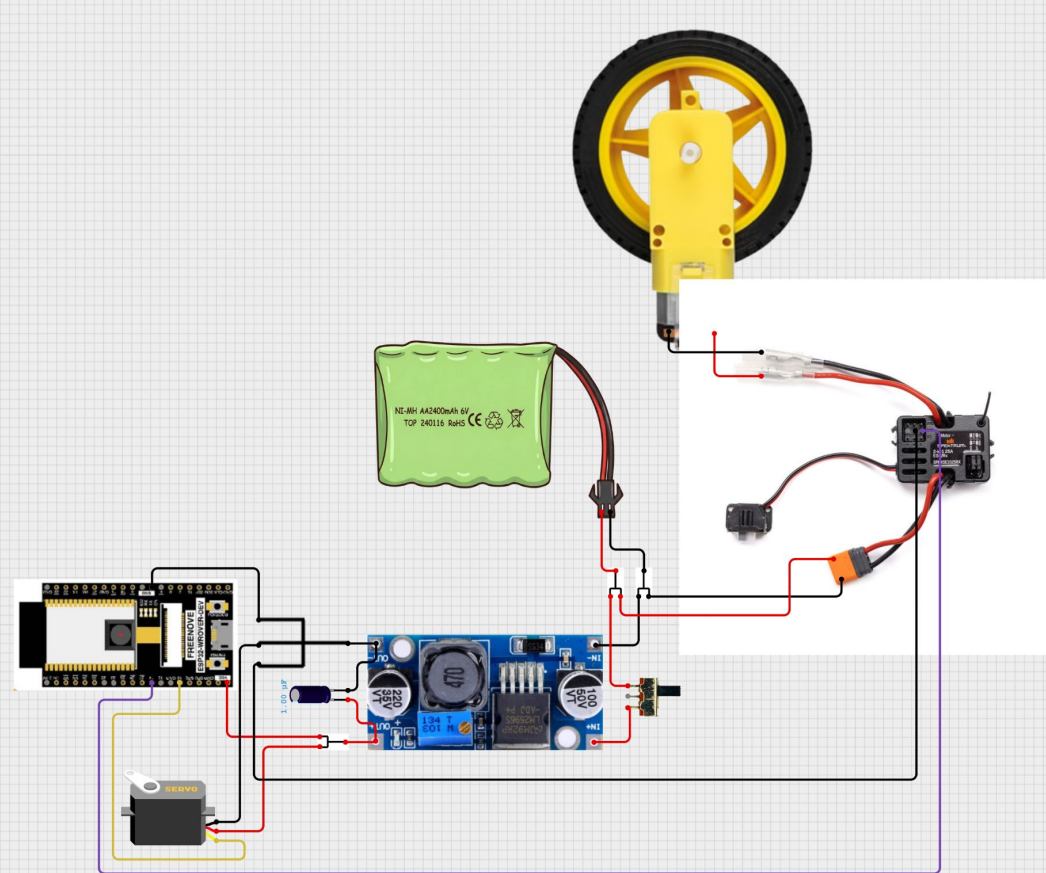
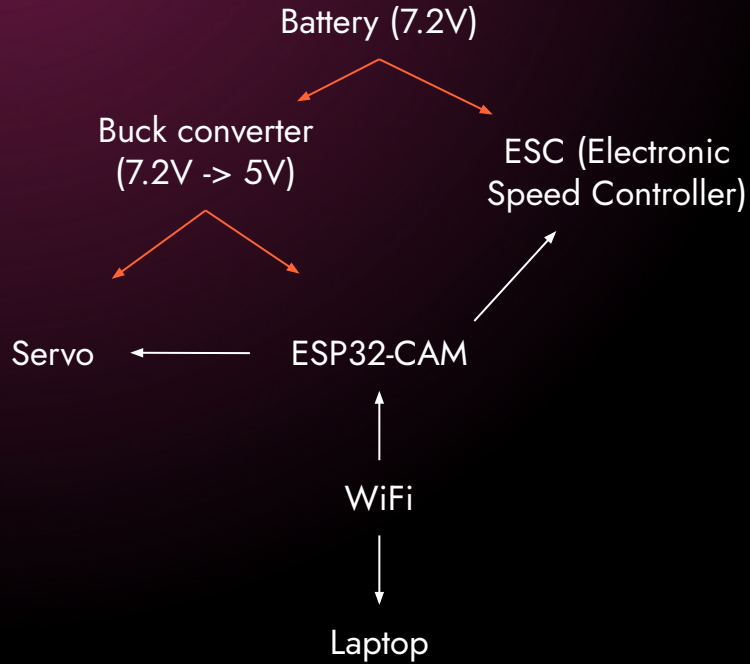


- Fully onboard AI processing
- Eliminate need for external laptop
- Real-time object detection directly on the vehicle



Object Detection Demonstration

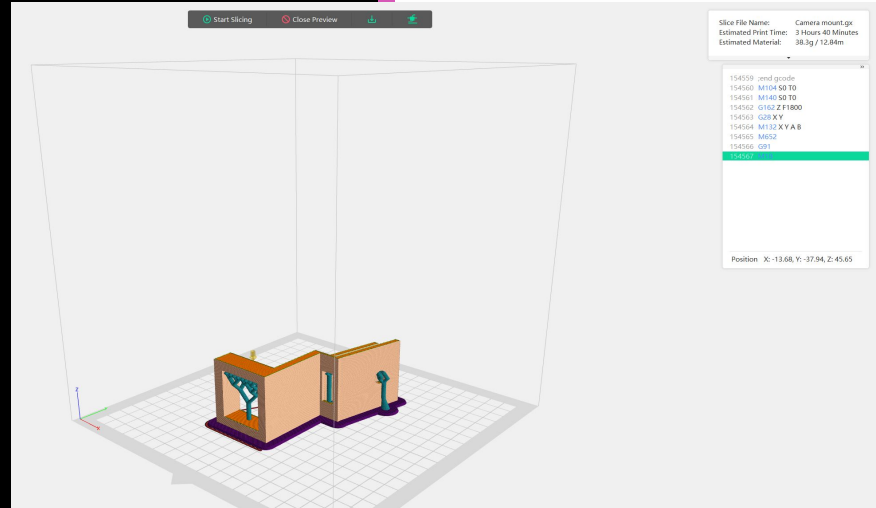
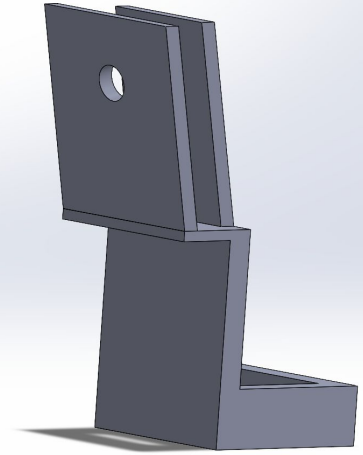
Hardware Plan



CAD Model for camera mounting

We have created a model for the mounting system of the ESP32 camera and board kit. This will contain our primary sensor.

- Mounts onto front bumper
- Board slides into the top
- Hole in the center for camera to see through
- Future plans to redesign model to have a spot for LIDAR sensor



Next Steps:

- Assemble electronics
- Steering and throttle control from laptop
- Establish microcontroller video stream on laptop
- Connect YOLO detections to navigation logic
- Preliminary autonomous testing
- Integrate Lidar sensor into Solidworks Model

THANK YOU

Any questions?